Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data columns tracked

37

Missing Values

Review ratings imputed

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Data Preparation Process

Data Loading

Imported with pandas

Exploration

Checked structure with df.info()

Cleaning

Handled missing values

Integration

Loaded to MySQL

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Feature Engineering Highlights

Age Groups

Binned customer ages for segmentation

Purchase Frequency

Calculated days between purchases

Column Standardization

Renamed to snake case for readability

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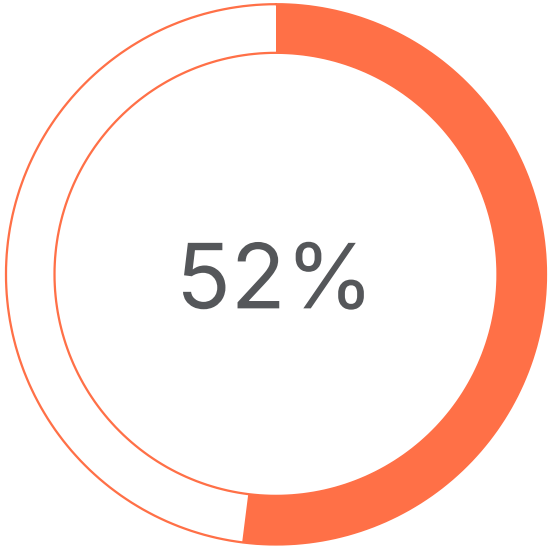
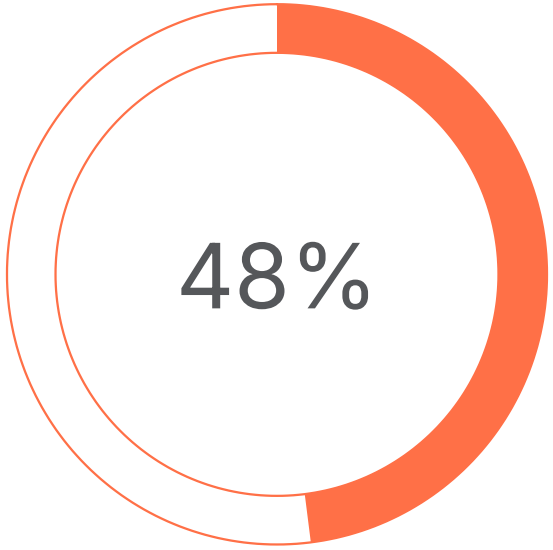
Revenue by Gender



Female Customers

Generated majority of revenue



Male Customers

Contributed nearly half of total sales

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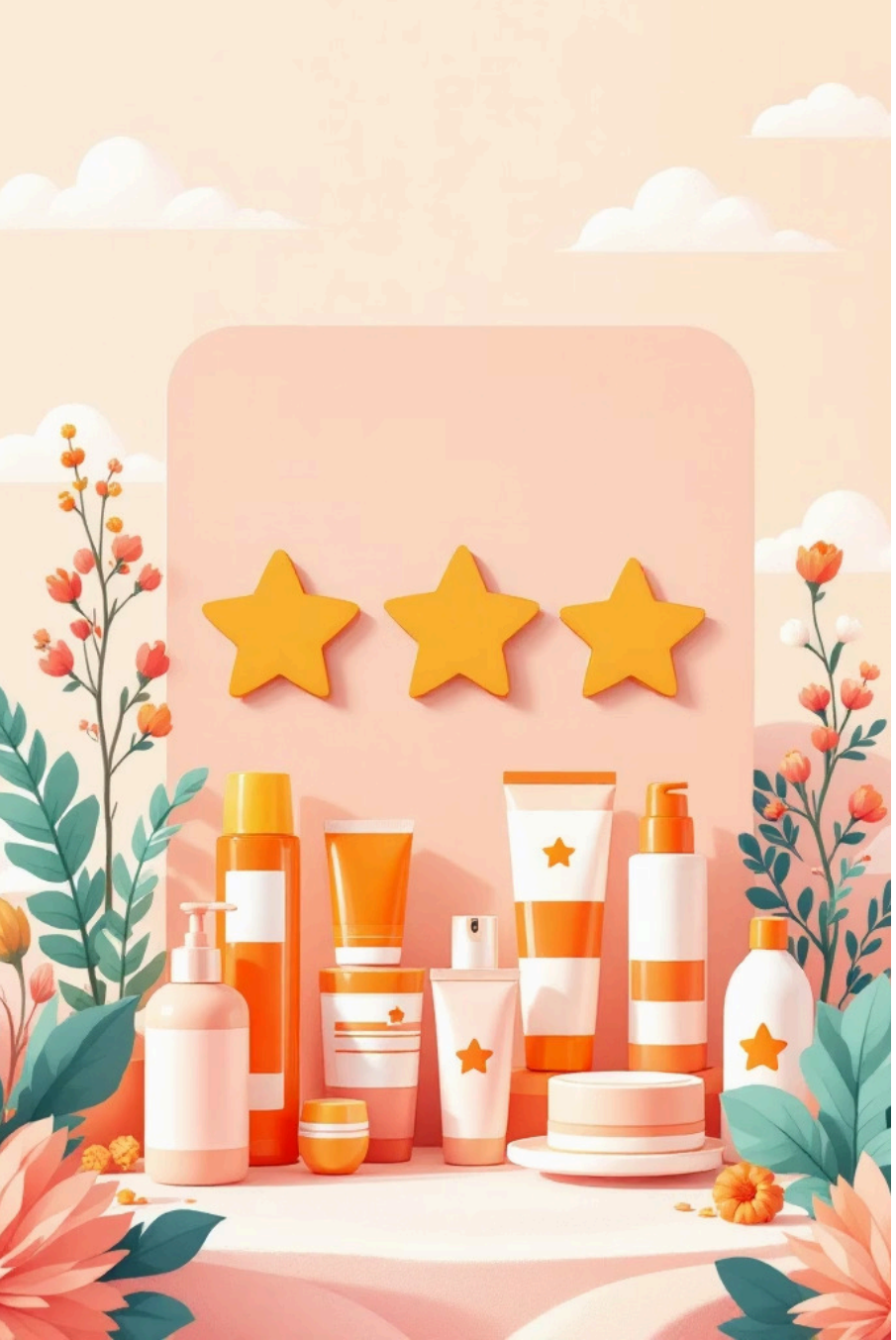

Top Products by Rating

Rated Products

Top 5 products identified by average review ratings

Discount Analysis

5 products with highest percentage of discounted purchases

Category Leaders

Top 3 products listed per category

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Shipping & Subscription Insights

Express Shipping

Higher average purchase amounts vs. standard shipping

Subscribers

Compared average spend and total revenue across subscription status

Repeat Buyers

Customers with >5 purchases more likely to subscribe

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Customer Segmentation

New Customers

First-time purchases identified

Returning Customers

Multiple purchases tracked

Loyal Customers

High-frequency buyers classified

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Revenue by Age Group

26.66%

Young Adult

Top revenue contributor

25.39%

Middle-Aged

Second highest revenue

24%

Adult

Steady spending pattern

23.92%

Senior

Emerging segment

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Strategic Recommendations

1

Boost Subscriptions

Promote exclusive benefits for subscribers

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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